STT 811 Project

```r
grocery <- read.csv("grocery_inventory_clean.csv")
```

```r
library(tidyverse)
```

```
## Warning: package 'tidyverse' was built under R version 4.3.3

## Warning: package 'dplyr' was built under R version 4.3.3

## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.0     v stringr   1.5.1
## v ggplot2   3.4.4     v tibble    3.2.1
## v lubridate 1.9.3     v tidyr     1.3.1
## v purrr     1.0.2
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
library(ggplot2)
library(ggiraph)
library(caret)
```

```
## Warning: package 'caret' was built under R version 4.3.3

## Loading required package: lattice
##
## Attaching package: 'caret'
##
## The following object is masked from 'package:purrr':
##
##     lift
```

```r
library(xgboost)
```

```
## Warning: package 'xgboost' was built under R version 4.3.3

##
## Attaching package: 'xgboost'
##
## The following object is masked from 'package:dplyr':
##
##     slice
```

```r
library(randomForest)
```

```
## Warning: package 'randomForest' was built under R version 4.3.3
```

```
## randomForest 4.7-1.2
## Type rfNews() to see new features/changes/bug fixes.
##
## Attaching package: 'randomForest'
##
## The following object is masked from 'package:dplyr':
##
##     combine
##
## The following object is masked from 'package:ggplot2':
##
##     margin
```

```r
library(nnet)
library(dplyr)
```

```r
grocery$FIFO_FEFO <- as.factor(grocery$FIFO_FEFO)
grocery$Inventory_Status <- as.factor(grocery$Inventory_Status)
grocery$ABC_Class <- as.factor(grocery$ABC_Class)
stock_variables <- c("Quantity_On_Hand", "Quantity_Reserved", "Quantity_Committed", "Damaged_Qty", "Retu
"Stock_Age_Days")
planning_variables <- c("Avg_Daily_Sales", "Forecast_Next_30d", "Days_of_Inventory", "Reorder_Point", "S
financial_variables <- c("Unit_Cost_USD", "Last_Purchase_Price_USD", "Total_Inventory_Value_USD", "Suppl
grocery_stock_status <- grocery[c(stock_variables, 'ABC_Class')]
grocery_planning_velocity <- grocery[c(planning_variables, 'ABC_Class')]
grocery_financial_quality <- grocery[c(financial_variables, 'ABC_Class')]
```

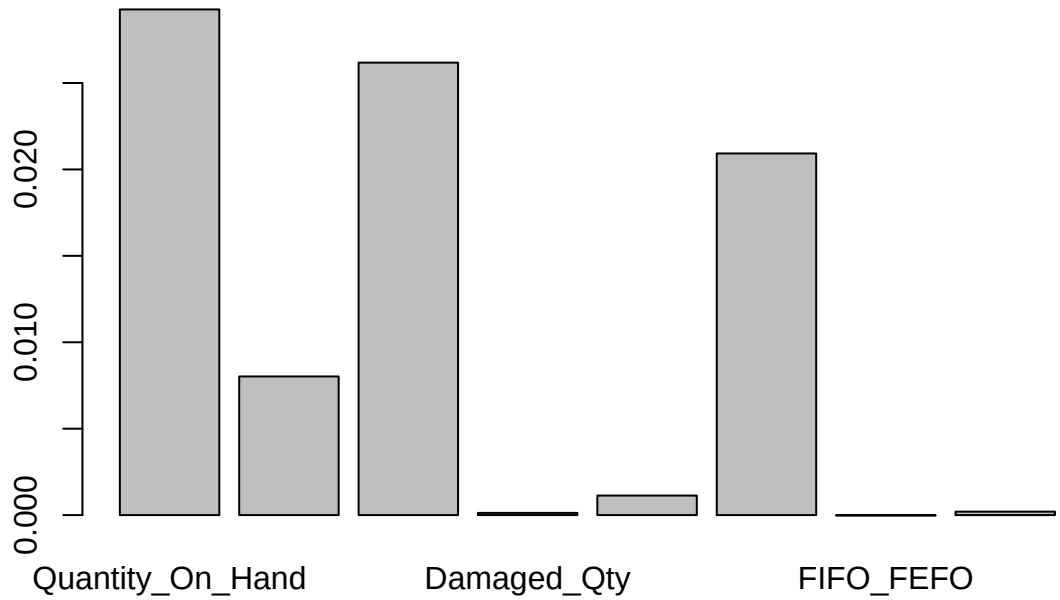
```r
split_pct <- 0.8
n <- length(grocery$ABC_Class)*split_pct
row_samp <- sample(1:length(grocery$ABC_Class), n, replace = FALSE)
train <- grocery[row_samp,]
test <- grocery[-row_samp,]
train_stock_status <- grocery_stock_status[row_samp,]
test_stock_status <- grocery_stock_status[-row_samp,]
train_planning_velocity <- grocery_planning_velocity[row_samp,]
test_planning_velocity <- grocery_planning_velocity[-row_samp,]
train_financial_quality <- grocery_financial_quality[row_samp,]
test_financial_quality <- grocery_financial_quality[-row_samp,]
```

```r
grocery_stock_rf <- randomForest(ABC_Class ~ . - ABC_Class,
                      data = train_stock_status, mtry = 2,
                      importance = TRUE, ntree = 1000, maxnodes = 4)
rf_predict_stock <- predict(grocery_stock_rf, test_stock_status)
confusionMatrix(rf_predict_stock, as.factor(test_stock_status$ABC_Class))
```

```
## Confusion Matrix and Statistics
##
```

```
##           Reference
## Prediction  A  B  C
##          A  0  0  0
##          B 12  2  5
##          C 34 58 89
##
## Overall Statistics
##
##                Accuracy : 0.455
##                  95% CI : (0.3846, 0.5267)
##     No Information Rate : 0.47
##     P-Value [Acc > NIR] : 0.6896
##
##                   Kappa : 0.0021
##
##  Mcnemar's Test P-Value : <2e-16
##
## Statistics by Class:
##
##                      Class: A Class: B Class: C
## Sensitivity              0.00  0.03333   0.9468
## Specificity              1.00  0.87857   0.1321
## Pos Pred Value            NaN  0.10526   0.4917
## Neg Pred Value           0.77  0.67956   0.7368
## Prevalence               0.23  0.30000   0.4700
## Detection Rate           0.00  0.01000   0.4450
## Detection Prevalence     0.00  0.09500   0.9050
## Balanced Accuracy        0.50  0.45595   0.5394
```

```r
barplot(grocery_stock_rf$importance[,3])
```

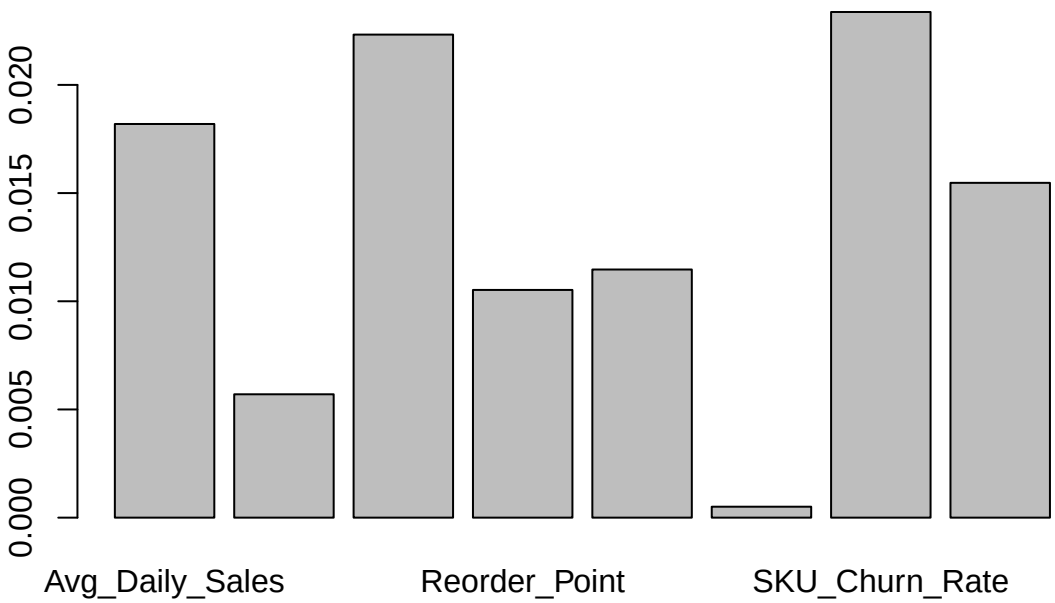



```
grocery_planning_rf <- randomForest(ABC_Class ~ . - ABC_Class,
                      data = train_planning_velocity, mtry = 2,
                      importance = TRUE, ntree = 1000, maxnodes = 4)
rf_predict_planning <- predict(grocery_planning_rf,  test_planning_velocity)
confusionMatrix(rf_predict_planning, as.factor(test_planning_velocity$ABC_Class))
```

```
## Confusion Matrix and Statistics
##
##           Reference
## Prediction  A  B  C
##          A  0  0  0
##          B  8  9  2
##          C 38 51 92
##
## Overall Statistics
##
##                Accuracy : 0.505
##                  95% CI : (0.4336, 0.5763)
##     No Information Rate : 0.47
##     P-Value [Acc > NIR] : 0.1785
##
##                   Kappa : 0.0937
##
##  Mcnemar's Test P-Value : <2e-16
##
```

```
## Statistics by Class:
##
##                      Class: A Class: B Class: C
## Sensitivity              0.00   0.1500   0.9787
## Specificity              1.00   0.9286   0.1604
## Pos Pred Value            NaN   0.4737   0.5083
## Neg Pred Value           0.77   0.7182   0.8947
## Prevalence               0.23   0.3000   0.4700
## Detection Rate           0.00   0.0450   0.4600
## Detection Prevalence     0.00   0.0950   0.9050
## Balanced Accuracy        0.50   0.5393   0.5696
```

```r
barplot(grocery_planning_rf$importance[,3])
```

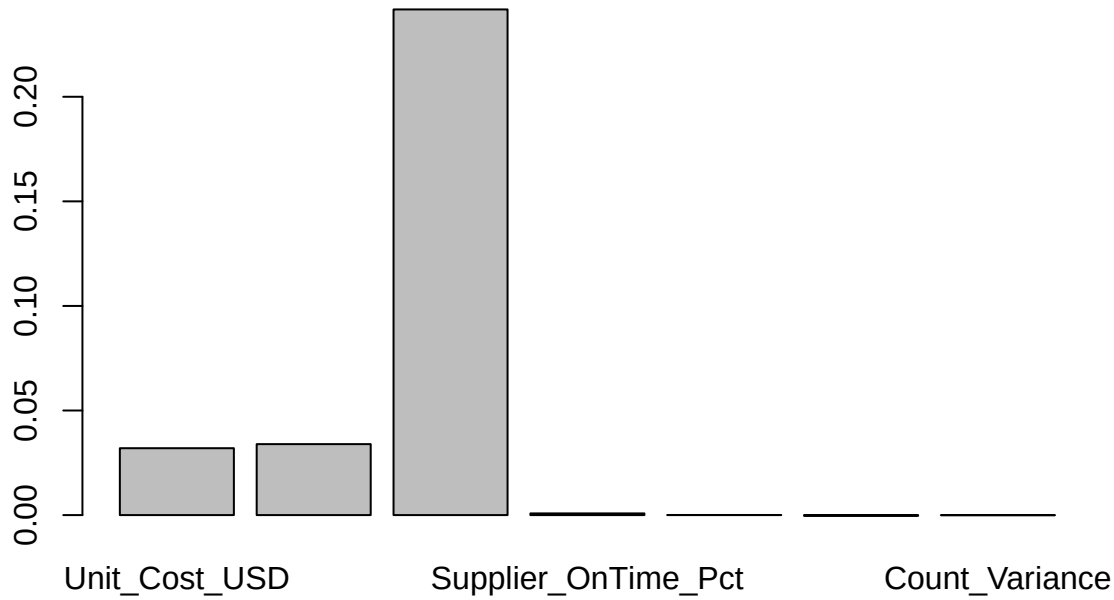



```r
grocery_financial_rf <- randomForest(ABC_Class ~ . - ABC_Class,
                      data = train_financial_quality, mtry = 2,
                      importance = TRUE, ntree = 1000, maxnodes = 4)
rf_predict_financial <- predict(grocery_financial_rf, test_financial_quality)
confusionMatrix(rf_predict_financial, as.factor(test_financial_quality$ABC_Class))
```

```
## Confusion Matrix and Statistics
##
##           Reference
## Prediction  A  B  C
##          A  2  0  0
```

```
##          B 39 60  0
##          C  5  0 94
##
## Overall Statistics
##
##                Accuracy : 0.78
##                  95% CI : (0.7161, 0.8354)
##     No Information Rate : 0.47
##     P-Value [Acc > NIR] : < 2.2e-16
##
##                   Kappa : 0.6432
##
##  Mcnemar's Test P-Value : NA
##
## Statistics by Class:
##
##                      Class: A Class: B Class: C
## Sensitivity           0.04348   1.0000   1.0000
## Specificity           1.00000   0.7214   0.9528
## Pos Pred Value        1.00000   0.6061   0.9495
## Neg Pred Value        0.77778   1.0000   1.0000
## Prevalence            0.23000   0.3000   0.4700
## Detection Rate        0.01000   0.3000   0.4700
## Detection Prevalence  0.01000   0.4950   0.4950
## Balanced Accuracy     0.52174   0.8607   0.9764
```

```r
barplot(grocery_financial_rf$importance[,3])
```

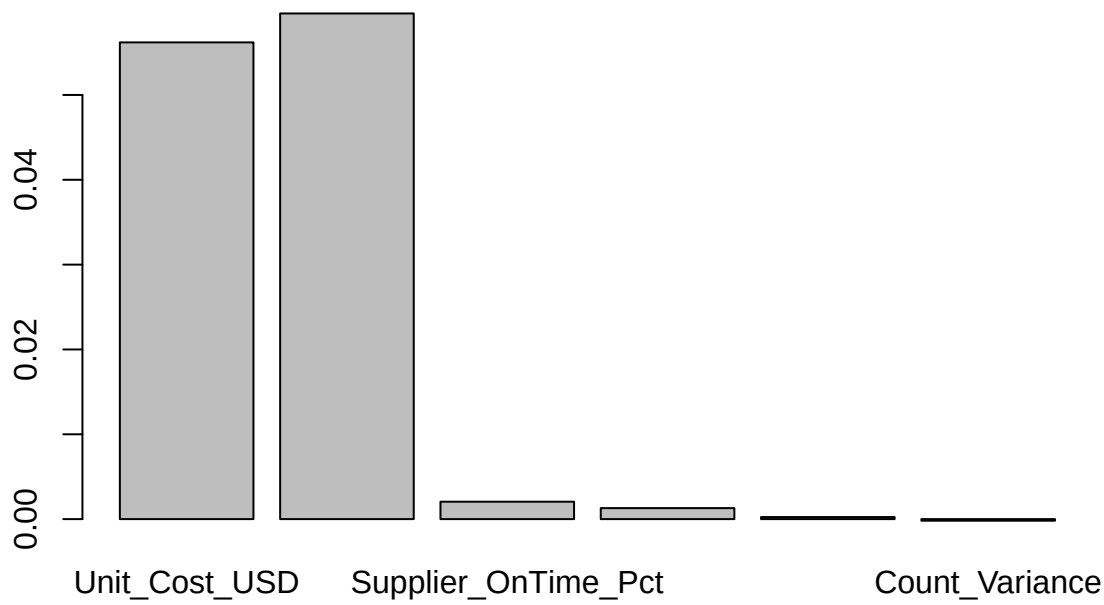



```
grocery_financial_rf <- randomForest(ABC_Class ~ . - ABC_Class - Total_Inventory_Value_USD,
                      data = train_financial_quality, mtry = 2,
                      importance = TRUE, ntree = 1000, maxnodes = 4)
rf_predict_financial <- predict(grocery_financial_rf, test_financial_quality)
confusionMatrix(rf_predict_financial, as.factor(test_financial_quality$ABC_Class))
```

```
## Confusion Matrix and Statistics
##
##           Reference
## Prediction  A  B  C
##          A  2  0  1
##          B 38 32 24
##          C  6 28 69
##
## Overall Statistics
##
##                Accuracy : 0.515
##                  95% CI : (0.4435, 0.5861)
##     No Information Rate : 0.47
##     P-Value [Acc > NIR] : 0.1143
##
##                   Kappa : 0.2095
##
##  Mcnemar's Test P-Value : 4.256e-09
##
```

```
## Statistics by Class:
##
##                      Class: A Class: B Class: C
## Sensitivity           0.04348   0.5333   0.7340
## Specificity           0.99351   0.5571   0.6792
## Pos Pred Value        0.66667   0.3404   0.6699
## Neg Pred Value        0.77665   0.7358   0.7423
## Prevalence            0.23000   0.3000   0.4700
## Detection Rate        0.01000   0.1600   0.3450
## Detection Prevalence  0.01500   0.4700   0.5150
## Balanced Accuracy     0.51849   0.5452   0.7066
```

```r
barplot(grocery_financial_rf$importance[,3])
```




```r
grocery_financial_rf <- randomForest(ABC_Class ~ Total_Inventory_Value_USD,
                      data = train_financial_quality, mtry = 2,
                      importance = TRUE, ntree = 1000, maxnodes = 4)
```

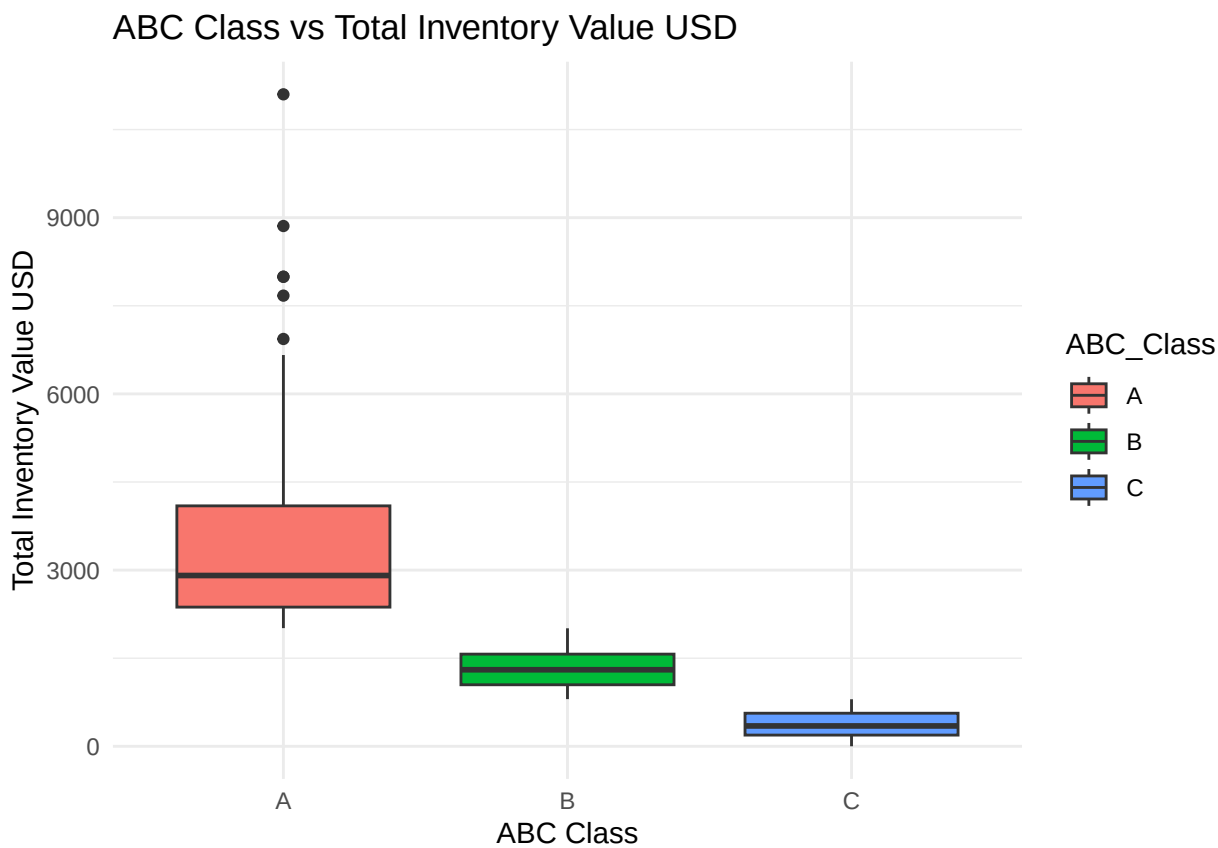
```
## Warning in randomForest.default(m, y, ...): invalid mtry: reset to within valid
## range
```

```r
rf_predict_financial <- predict(grocery_financial_rf, test_financial_quality)
confusionMatrix(rf_predict_financial, as.factor(test_financial_quality$ABC_Class))
```

```
## Confusion Matrix and Statistics
##
##           Reference
## Prediction  A  B  C
##          A 46  0  0
##          B  0 60  0
##          C  0  0 94
##
## Overall Statistics
##
##                Accuracy : 1
##                  95% CI : (0.9817, 1)
##     No Information Rate : 0.47
##     P-Value [Acc > NIR] : < 2.2e-16
##
##                   Kappa : 1
##
##  Mcnemar's Test P-Value : NA
##
## Statistics by Class:
##
##                      Class: A Class: B Class: C
## Sensitivity              1.00      1.0     1.00
## Specificity              1.00      1.0     1.00
## Pos Pred Value           1.00      1.0     1.00
## Neg Pred Value           1.00      1.0     1.00
## Prevalence               0.23      0.3     0.47
## Detection Rate           0.23      0.3     0.47
## Detection Prevalence     0.23      0.3     0.47
## Balanced Accuracy        1.00      1.0     1.00
```

```r
#barplot(grocery_financial_rf$importance[,3])
```

```r
ggplot(grocery, aes(x = ABC_Class, y = Total_Inventory_Value_USD, fill = ABC_Class)) +
  geom_boxplot() +
  labs(title = "ABC Class vs Total Inventory Value USD",
       x = "ABC Class",
       y = "Total Inventory Value USD") +
  theme_minimal()
```



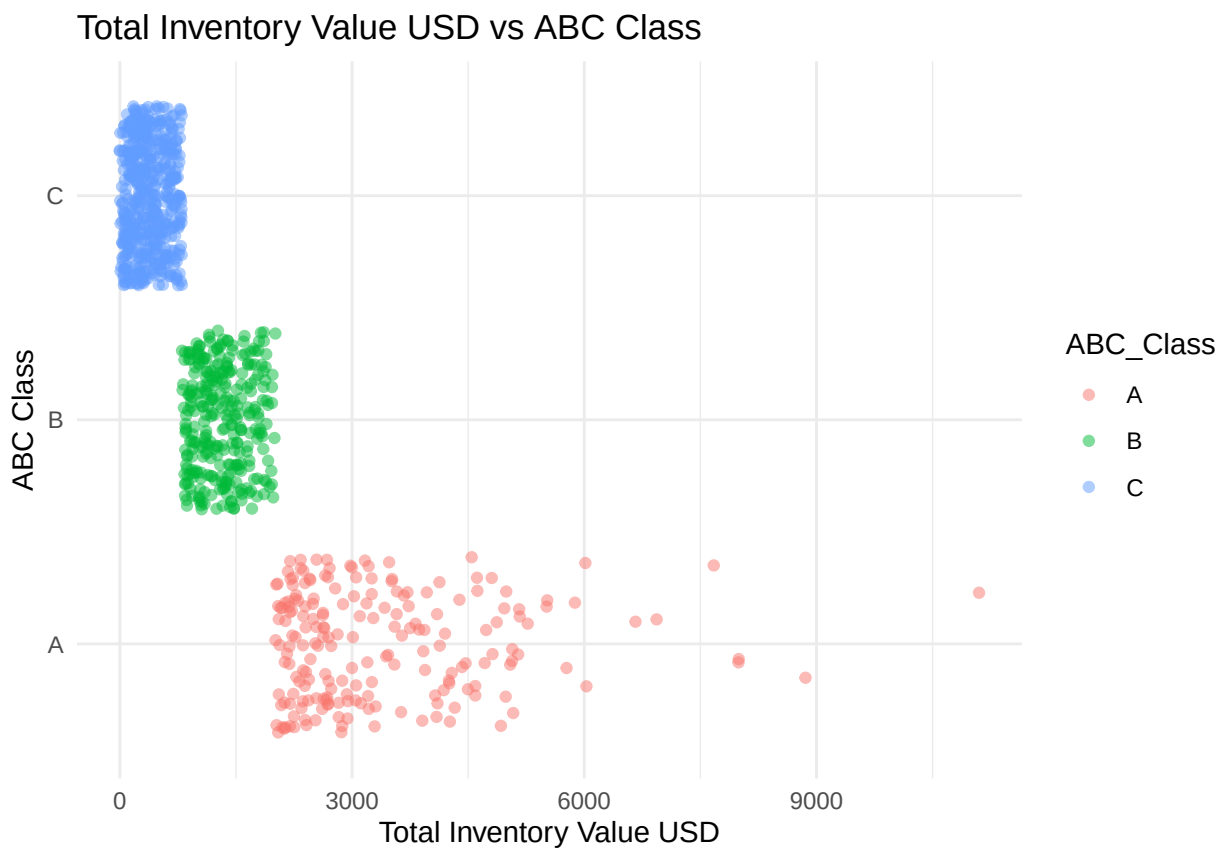

```r
model_logit <- multinom(
  ABC_Class ~ Total_Inventory_Value_USD,
  data = train)
```

```
## # weights:  9 (4 variable)
## initial  value 878.889831
## iter  10 value 102.982602
## iter  20 value 15.762164
## iter  30 value 10.736939
## iter  40 value 8.853994
## iter  50 value 7.339323
## iter  60 value 6.061138
## iter  70 value 5.538980
## iter  80 value 5.253113
## iter  90 value 5.133936
## iter 100 value 4.971878
## final  value 4.971878
## stopped after 100 iterations
```

```r
pred_logit <- predict(model_logit, test)
mean(pred_logit == test$ABC_Class)
```

```
## [1] 1
```

```r
ggplot(grocery, aes(x = Total_Inventory_Value_USD, y = ABC_Class, color = ABC_Class)) +
  geom_jitter(alpha = 0.5) +
    labs(title = "Total Inventory Value USD vs ABC Class",
       y = "ABC Class",
       x = "Total Inventory Value USD") +
  theme_minimal()
```




```r
grocery %>%
  group_by(ABC_Class) %>%
  summarise(
    min_value = min(Total_Inventory_Value_USD, na.rm = TRUE),
    max_value = max(Total_Inventory_Value_USD, na.rm = TRUE))
```

```
## # A tibble: 3 x 3
##   ABC_Class min_value max_value
##   <fct>         <dbl>     <dbl>
## 1 A             2012.    11100
## 2 B              803.     2009.
## 3 C                0       801.
```